Haptic Belt with Pedestrian Detection

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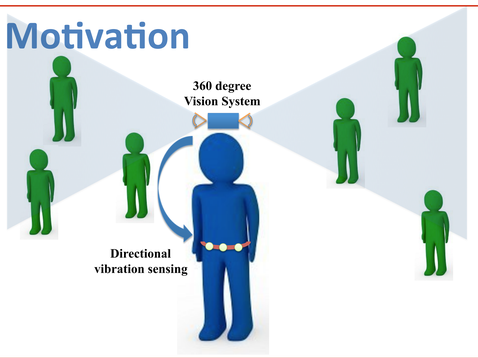
1 Abstract

We built a haptic belt that sends vibrating signals to inform the user of the positions of surrounding pedestrians. When the camera detects a person, a motor in that direction will vibrate. By wearing the belt, the user can develop a "sixth sense" to identify where people are standing, even if a person stands behind the user. Our belt improves upon earlier work in wearable haptics by transmitting only high-level processed data to users.

2 Motivation



- Utilizing the sense of touch, haptic devices have been described as "**doing for the sense of touch what computer graphics for vision**"[3].
- In highly pressurized environments, **human visual attention is indispensible and often overloaded**; haptic support is useful in reducing vision attention.
- Computer vision can help give us 360 degrees field of view: '**Having eyes at the back**'
- **Our system transfers the ability to 'see' people in 360 degrees field of view to haptic vibrations**, by combining a state-of-the-art computer vision system with a haptic belt

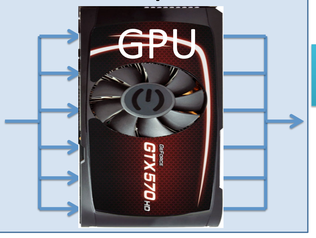
3 System Overview

CPU: Intel i7 2600K 3.4GHz

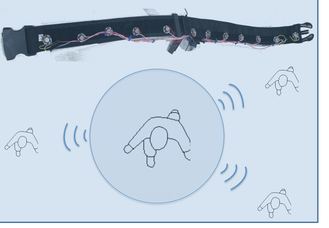
GPU: NVIDIA GTX 570
cores : 480
Graphics Clock:732 MHz
Processor Clock :1464 MHz
Memory: 1280 MB

8 Cameras covering 360 degrees of view

Convolutional Network Detector running at 1.5 seconds/frame



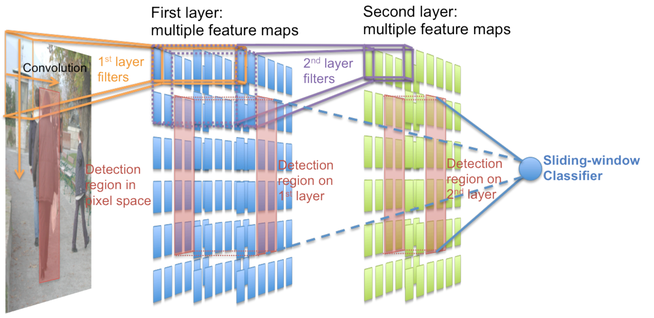
Haptic belt 12 vibrating motors to indicate direction



4 Convolutional Neural Networks

For the human detector we employ convolutional neural networks [5]:

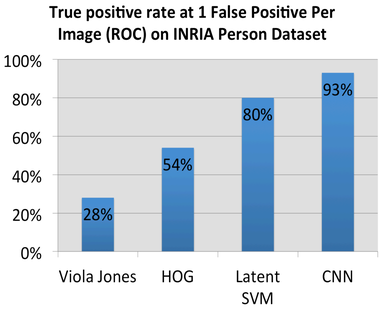
- Multi-stage [1] architectures for visual recognition
- Few parameters due to weight-sharing
- All parameters can be tuned with back-propagation



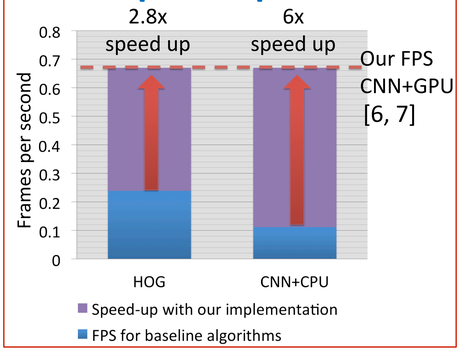
5 State-of-the-art Detector

The system applies pedestrian detector by Pierre et. al. [5] using convolutional neural networks implemented in the C++ EBLearn framework [4]

As shown on the right, CNN achieves state-of-the-art results on INRIA Person Dataset [5]



6 GPU Speed-up



7 References

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[2] Koray Kavukcuoglu, Pierre Sermanet, Y-Lan Boureau, Karol Gregor, Michael Mathieu, Yann LeCun. Learning Convolutional Feature Hierarchies for Visual Recognition. NIPS 2010.
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[4] Pierre Sermanet Koray Kavukcuoglu Yann LeCun. EBLearn: Open-Source Energy-Based Learning in C++. ICTAI 2009
[5] Pierre Sermanet, Koray Kavukcuoglu and Yann LeCun .Traffic Signs and Pedestrians Vision with Multi-Scale Convolutional Networks. Snowbird Workshop 2011
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